

Panel: OCP Storage Specifications Update and What Customers Care About

David Black *Senior Distinguished Engineer, Dell*

Chris Sabol *Storage Architect, Google*

Jeff Wolford *Distinguished Technologist, HPE Server Options*

Vineet Parekh *Hardware Systems Engineer, Meta*

Tim Sharp *Principal Hardware Engineer, Microsoft*

OCP Datacenter Specifications Updates

2026: Datacenter NVMe SSD Specification (only)



OCP GLOBAL SUMMIT | 2026

Datacenter NVMe SSD Specification v2.8 Update

- Authors: Dell, Google, HPE, Meta, Microsoft
- OCP NVMe v2.8 Specification is complete
- Contribution in Progress
  - Publication ETA: November or early December



Datacenter NVMe® SSD Specification

Version 2.8r31 (08202026)

Author: Vineet Parekh, Meta
Author: Tim Sharp, Microsoft
Author: Jeff Wolford, TJ Choi, HPE
Author: David Black, Colm Murphy, Dell Technologies
Author: Chris Sabol, Charles Kunzman, Google

Improvements (1 of 2)

- Error recovery: Workflows rewritten for consistent host behavior
  - Minor impacts on Error Recovery Log Page (C1h)
- Event injection: Framework revised for consistent device behavior
  - Focus Areas: Notifications and exiting/ending an injected event
  - Also added some more events to inject
- Human Readable (via nvme-cli) functionality scope expanded:
  - Vendor-specific telemetry (Data Areas 3 and 4), Latency Monitor data
- CNSA 2.0 Post-Quantum Cryptography (PQC): Now required
  - Dual firmware image signatures: sign with both CNSA 1.0 and CNSA 2.0 signatures
  - SPDM 1.4.1, including dual-bank for certificates, with SPDM 1.1 backwards compatibility

Improvements (2 of 2)

- TCG Locking – New requirements:
  - Non-global locking range support, use up to 16 locking ranges with same namespace
  - Firmware download port locking
- Admin Command History: New log page + Feature to configure events to log
  - Intended usage: Host software debugging
- Device personality support for BMaaS (Bare Metal as a Service):
  - Added to profile B (enterprise servers)
- PCIe Uncorrectable Errors:
  - New running total SMART metric (counter)
  - New Feature to clear underlying PCIe Uncorrectable Error counters
- Numerous other minor changes and editorial cleanup

Specification Revision Plans

- OCP Datacenter SAS-SATA Spec was not revised in 2026.
  - Every-other-year cadence for this specification is a distinct possibility.
- Tentative technical activity schedule:
  - Nov-Dec 2026: Initial OCP NVMe 2.9 spec revisions
  - Jan-Mar 2027: Revise OCP SAS-SATA spec
  - Apr-Sep 2027: Complete OCP NVMe 2.9 spec
    - Stretch Goal: finish by end of August, if possible.

A big shoutout to all of you who signed the Feedback CLA, sent the authors feedback and participated in the monthly feedback sessions!



Panel: What Customers Care About

David Black, Dell

Chris Sabol, Google

Jeff Wolford, HPE

Vineet Parekh, Meta

Tim Sharp, Microsoft



Dell

- Common OCP Spec
  - Avoid pointless “Have One Built for You” divergence
  - Industry baseline greatly increases likelihood that drive “just works”
- OCP NVMe Event Injection
  - Test host & drive responses before drives misbehave
  - Injected event behavior specified in detail
- OCP NVMe Error Recovery
  - Automatic recovery is often possible, even for some hardware problems
  - Avoid support expenses for unlikely to repeat firmware crash/assert
  - Extracted error information supports offline analysis

Dell (continued)

- Direct Liquid Cooling (cold plates)
  - Common industry approach is crucial
  - Emerging development: liquid-cooling-only data centers
- Bare Metal as a Service
  - Prevent tenants from causing damage
  - Example: block tenant attempts to update drive firmware.
- Post Quantum Crypto (CNSA 2.0)
  - Firmware image signatures
  - Drive attestation (SPDM 1.4.1)
  - Pay attention to transition considerations

Google

- Security
  - Challenge: Known vulnerability on one SSD is a presumed risk on all SSDs
  - Solution:
    - Caliptra trusted open RTL and firmware
    - Attestation through SPDM
    - OCP LOCK trusted key management; exposed through TCG MEK-MPA extension
    - OCP SAFE drive audits
- Performance
  - Challenge: Preventing performance and especially latency issues
  - Solution:
    - NVMe rate limiting (TP4176 in NVMe base spec v2.4)
    - NVMe IO prioritization (TP4221)

Google (continued)

- Debuggability
  - Challenge: Debugging complex issues in production
  - Solution:
    - OCP human readable telemetry
    - OCP latency monitoring
    - Commitment by Storage Spec authors and SSD vendors to continue improving over time
- Bare Metal
  - Challenge: Out of band telemetry and SSD management require more bandwidth
  - Solution:
    - SPDM and NVMe-MI provide the required interfaces
    - Move to MCTP over USB for out of band

HPE

- Security
  - Dual (Classical and CNSA 2.0) signatures at all levels
    - Hardware Root of Trust (HW RoT)
    - Secure boot
    - secure firmware update
    - SPDM v1.2 (Classic) and v1.4.1 (PQC) attestation
  - Cryptographic agility to support future additional PQC algorithms and lifecycle security
- OOB (Out of Band) support
  - Adding I3C for higher transfer rates to support PQC signatures (e.g., SPDM v1.4.1 attestation)
- NVMe Mainstream / Datacenter class for SATA and SAS SSD replacement
  - maintain lowest capacity options for boot and cost optimizations

HPE (continued)

- M.2
  - Low capacity (480 / 960 / 1920 / 3840 GB)
  - SRIS (Separate Reference Clock Independent Spread) support
  - LED support as close to E3.S drives as possible
- Sustainability
  - Support OCP Data Center Power State command
  - Power efficient designs higher data rate while minimizing increasing power envelope
    - PCIe generation transitions are happening at an increasing cadence

Meta

- Common Specification
  - Alignment on cloud NVMe SSD features and specifications
- Flexible Data Placement
  - Reduced Write Amplification Factor (WAF) and improved drive endurance
- Latency Monitoring
  - Fine-grained I/O latency measurement and QoS tail latency tracking
- Structured Telemetry Logs
  - Standardized crash dump diagnostics and telemetry reporting

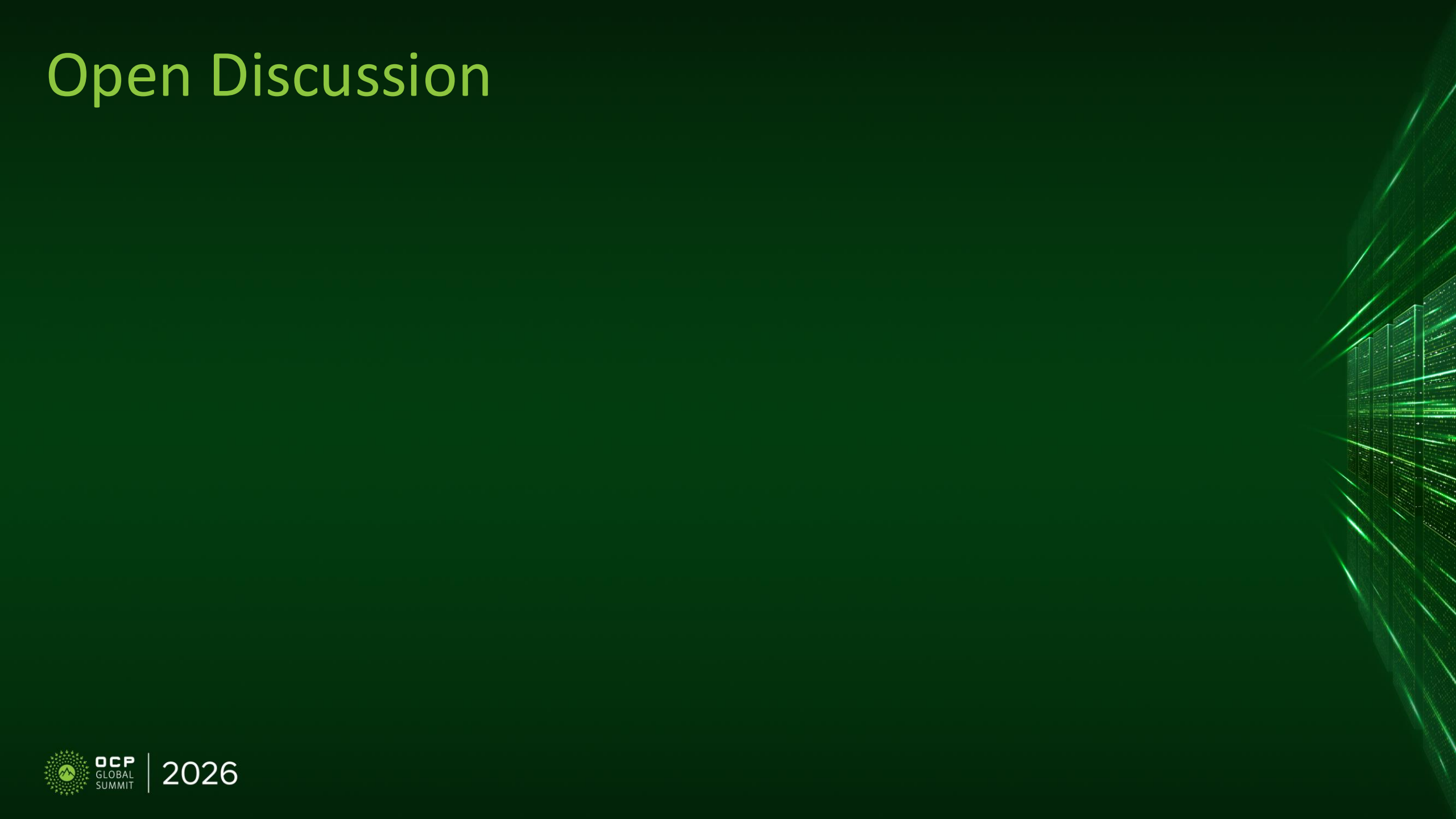
Microsoft

- Security Architecture Enhancements
  - Support for Caliptra silicon Root of Trust and device attestation
    - Secure firmware updates via streaming boot
  - Support for OCP-LOCK
    - Key management support
  - Support for SPDM 1.4.1
    - Platform device authentication/attestation
    - CNSA 1.0/2.0 (PQC)
      - Dual firmware image Signing
- Support for Enhanced Error Testing And Recovery
  - Impact-less firmware asserts
  - Automatic error recovery without error recovery workflow
  - Event injection testing

Microsoft (continued)

- OOB Support
  - I3C
    - Much higher transfer rates possible for all NVMe-MI based transactions
  - Future USB Support
    - Eventually even higher transfer rates for all NVMe-MI based transactions
- Controller Virtualization
  - Future adoption of industry standard NVMe virtualization proposals
    - SIOV
    - Exported NVM subsystems
    - PCIe Infrastructure for Live Migration

Open Discussion



Thank You!

